ZX48 POWER AND TEST POINTS

ZX Spectrum 48K. Check the supply and the rails before suspecting any chip. Underlined pins open the drawing.

PSU SUPPLY

SINCLAIR ZX POWER PACK (9 V DC UNREGULATED) EXTERNAL

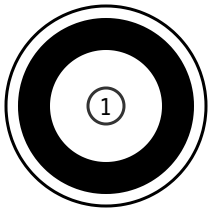
9 V DC unregulated, 1.4 A

KNOWN PROBLEMS

- CENTRE NEGATIVE: centre pin 0 V, sleeve +9 V, into a 2.1 mm coaxial socket (k1; the Sinclair manuals do not state the polarity). A centre-positive adapter puts the supply in backwards.
- The pack is a mains transformer, full-wave rectifier and smoothing capacitor with a thermal fuse in the primary; nothing is regulated (service manual 1.11). Design limits: at least 7.0 V at 1.4 A on 215 V AC, under 14 V peak at 600 mA on 265 V AC (5.2). About 15 V with no load (k1).
- Draw from a 9 V bench supply: 500-700 mA for a 16K, 700-900 mA for a 48K (3.1.4). Much more means a short; set the current limit just above that.
- Regulator input (C50 +ve) should read +9 V ±2 V; below +7 V the 7805 does not regulate (5.3.1). A replacement 7805 needs thermal compound and a firm bolt to the heatsink, kept clear of the edge connector (7.2).
- The DC-DC converter TR4/TR5 makes +12 V and -5 V for the 4116s. Smoke at switch-on or a dead board is often TR4 shorted base to collector: remove TR4, check TR5 and C44, and change TR4 and TR5 together if TR4 blows again (6.1).
- Check the rails at IC6 before any RAM swap: pin 9 +5 V ±0.25 V, pin 1 -5 V (-4 V to -5.5 V), pin 8 +12 V ±1.2 V (3.1.2). Mostek: the 4116 Vbb (-5 V) goes on first and off last and must never rise above 0 V while +12 V is applied, so a failed -5 V rail can kill the whole lower bank. Fix the converter before fitting new 4116s.
- Issue 4B and later boards carry the DC-DC converter modification. The service manual supplement says its own printed modification steps are wrong: do not copy them (9.2, 9.6).

J1 POWER IN (9V DC) 2.1 MM COAXIAL DC SOCKET

DC barrel jack, centre 1, shell 2.



1 0 V

2 +9 V ±2 V on…

1 **0V** `0 V`. Centre pin: 0 V (negative). *(unverified)*

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2 **+9V** `+9 V ±2 V on load (C50); about 15 V with no load (k1)`. Sleeve: unregulated +9 V to the 7805 and the DC-DC converter. *(unverified)*

CONNECTOR SOURCES

ZX Spectrum Introduction manual, power supply, p. 7

ZX Spectrum Service Manual 8.7 Connectors (PWR 2.1mm co-axial socket), p. 44

Sinclair ZX Archive, ZX Spectrum Connectors (centre pin Gnd)

Checked against: ZX Spectrum Introduction manual p. 7 (9 V DC 1.4 A); Service Manual 8.7 (2.1 mm co-axial socket); polarity from k1

SOURCES

ZX Spectrum Introduction manual, power supply 9 V DC 1.4 A, p. 7

ZX Spectrum Service Manual 1.11 Power supplies, p. 15

ZX Spectrum Service Manual 5.2 Power supply unit, p. 25

ZX Spectrum Service Manual 6.1 TR4 faults, p. 30

ZX Spectrum Service Manual 9.2 DC-DC converter modification, p. 48

Mostek MK4116 data sheet, supply sequencing, PDF p. 107

Sinclair ZX Archive, ZX Spectrum Connectors (polarity, no-load voltage)

RAILS

TP1	**+9 V DC in** C50 positive side; spectrum-edge pin A4 Below +7 V the 7805 does not regulate.	+9 V ±2 V (7 to 11 V)
TP2	**+5 V** C34 positive side; 4116 pin 9 (IC6); ULA pin 14; Z80A pin 11; spectrum-edge pin A3 The ULA and the Z80 both draw from +5 V. Switch-mode drop-ins for the 7805 (Traco, Murata) need no heatsink; on an Issue 2 board in a rubber-key case only a horizontal-mount version fits (Retroleum).	+4.75 to +5.25 V, no visible ripple
TP3	**+12 V** DC-DC converter (D5/C44); 4116 pin 8 (IC6); spectrum-edge pin B22 Made by TR4/TR5, not the 7805. Out-of-range converter rails can damage the 4116s (Retroleum).	+10.8 to +13.2 V
TP4	**+12 VA** LT side of C52, to IC14 (LM1889) only Filtered by R62/C45 for the colour circuits.	+10.8 to +13.2 V
TP5	**-5 V** DC-DC converter; 4116 pin 1 (IC6); spectrum-edge pin B20 Out-of-range converter rails can damage the 4116s (Retroleum).	-4 V to -5.5 V
TP6	**GND** Z80A pin 29; ULA pin 40; 4116 pin 16; 23128 pin 14 (ROM IC5) Common return.	0 V

TEST POINTS

TP7 Crystal (14 MHz)

ULA pin 39 (OSC)

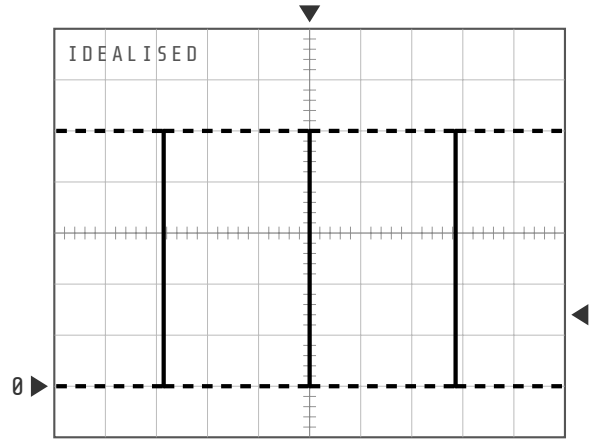
About 14 MHz. No clock, no picture.

TP8 CPU clock

ULA pin 32 (CLK), TR3, then Z80A pin 6

3.5 MHz square wave at full 5 V amplitude, with gaps during the display.

– SCOPE 100 ns/div · 1 V/div



IDEALISED FROM DATASHEET; DUTY NOT IN SOURCE

TIME/DIV	100 ns 3.5 cycles of 286 ns across 10 div
VOLTS/DIV	1 V 0 V to 5 V = 5 div
COUPLING	DC 0 V one division up
TRIGGER	Rising, 1.4 V TTL threshold
DUTY	— duty not in source
PROBE	10× channel set to 10×

Shape and level from the service manual, rate from the 14 MHz crystal divided by 4 (WoS 48K reference: 3.5 MHz). Low amplitude: check R25, then TR3 (6.3.2). Short stops while the ULA reads the screen are normal. The 5.3.1 table prints "14MHz" on this row; 6.3.2 gives 3 MHz at IC1 pin 32 and 14 MHz at pin 39.

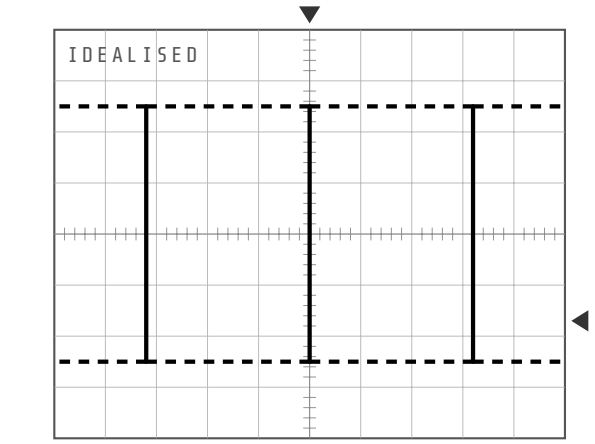
ZX Spectrum Service Manual 5.3.1 Basic checks (clock pulses: square wave at +5 V amplitude), p. 26

TP9 Luminance and sync (Y)

ULA pin 17

Line-rate pulses between about +1 V and +3.5 V, one line every 64 µs.

– SCOPE 20 µs/div · 500 mV/div



IDEALISED FROM CAPTURE; DUTY NOT IN SOURCE

TIME/DIV	20 µs 3.1 cycles of 64 µs across 10 div
VOLTS/DIV	500 mV 1 V to 3.5 V = 5 div
COUPLING	DC 0 V off screen: offset −2.25 V
TRIGGER	Rising, 1.4 V TTL threshold
DUTY	— duty not in source
PROBE	10× channel set to 10×

Service-manual drawing: +1 V base, stepping to +3.5 V each line. The drawing marks the period "65nS"; the 64 µs line period here is from the WoS 48K reference (224 T-states, see Clocks). Missing or flat with +5 V present: change the ULA.

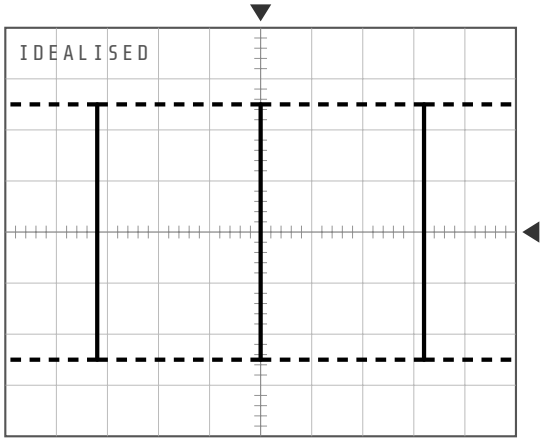
ZX Spectrum Service Manual 6.2.4 No video input at IC14 pin 12: IC1 pins 15, 16 and 17 waveforms, p. 31

TP10 Colour difference (U)

ULA pin 15

Rests at +2 V with short +3 V pulses each line.

– SCOPE 20 µs/div · 200 mV/div



IDEALISED FROM CAPTURE; DUTY NOT IN SOURCE

TIME/DIV	**20 µs** 3.1 cycles of 64 µs across 10 div
VOLTS/DIV	**200 mV** 2 V to 3 V = 5 div
COUPLING	**DC** 0 V off screen: offset –2.5 V
TRIGGER	**Rising, 2.5 V** mid-swing
DUTY	— duty not in source
PROBE	**10×** channel set to 10×

Service-manual drawing; line period from the WoS 48K reference (see Clocks). Wrong here but +5 V good: ULA. Right here but no colour: VR1/VR2 on Issue 1 and 2 boards, then IC14.

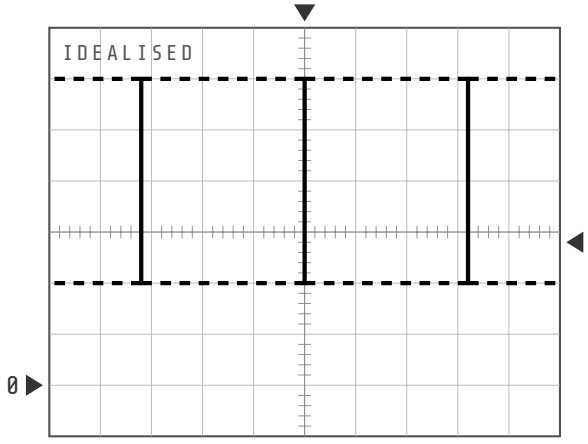
ZX Spectrum Service Manual 6.2.4, IC1 pin 15 waveform, p. 31

TP11 Colour difference (V)

ULA pin 16

Rests at +2 V with a pulse up to +3 V and a dip to +1 V each line.

– SCOPE 20 µs/div · 500 mV/div



IDEALISED FROM CAPTURE; DUTY NOT IN SOURCE

TIME/DIV	**20 µs** 3.1 cycles of 64 µs across 10 div
VOLTS/DIV	**500 mV** 1 V to 3 V = 4 div
COUPLING	**DC** 0 V one division up
TRIGGER	**Rising, 1.4 V** TTL threshold
DUTY	— duty not in source
PROBE	**10×** channel set to 10×

Service-manual drawing: +2 V rest, +3 V pulse, then a dip to +1 V later in the line; line period from the WoS 48K reference (see Clocks). The app draws a plain pulse between the two extremes.

ZX Spectrum Service Manual 6.2.4, IC1 pin 16 waveform, p. 31

TP12 Interrupt

ULA pin 12 (/INT)

Pulses LOW once per frame, every 20 ms.

TP13 Idle control lines

Z80A pins 17, 24, 25

/NMI, /WAIT and /BUSRQ HIGH.

TP14	**Reset** Z80A pin 26 (/RESET)	LOW at power-on, then HIGH.
TP15	**Lower RAM strobes** ULA pins 35, 1	/RAS and /CAS pulse LOW.
TP16	**Upper RAM strobes** 4532 pins 4, 3, 15 (IC22)	/RAS, /WE and /CAS pulse LOW on a 48K.
TP17	**Beeper** ULA pin 28 (SOUND)	Toggles when a BEEP runs.
TP18	**ULA temperature** ULA body	Warm is normal; too hot to hold for long is a fault in the making. Socket it.

CLOCKS

CLOCK	WHERE	FREQUENCY	SOURCE
Crystal X1	ULA pin 39	14 MHz	ZX Spectrum Service Manual 1.6 (external 14 MHz crystal X1), p. 13
Pixel clock *(unverified)*	Inside the ULA (no pin)	7 MHz	World of Spectrum, 48K reference: one pixel every half T-state
CPU clock *(unverified)*	ULA pin 32 to Z80A pin 6	3.5 MHz	World of Spectrum, 48K reference: 3.5 MHz / 69888 T-states per frame
Line rate *(unverified)*	ULA pin 17 (sync in Y)	15.625 kHz	World of Spectrum, 48K reference: 224 T-states per line (64 µs)
Frame interrupt *(unverified)*	ULA pin 12	50.08 Hz	World of Spectrum, 48K reference: 3.5 MHz / 69888 = 50.08 Hz
Colour subcarrier X2	IC14 (LM1889) leg 17	4.433619 MHz	ZX Spectrum Service Manual 3.1.4 Sub-carrier oscillator (4.433619 MHz ±50 Hz), p. 18

VIDEO LEVELS › PAL and NTSC line and sync timing

SOURCES

ZX Spectrum Service Manual 3.1.2 Voltage check, p. 17

ZX Spectrum Service Manual 5.3.1 Basic checks, p. 26

ZX Spectrum Issue 3 circuit diagram

ZX Spectrum Service Manual (scans), appended repair notes 3 and 5 (no coil buzz from inverter), PDF p. 64

Retroleum, Repairing a ZX Spectrum

Retroleum, Spectrum chips (switch-mode 7805 replacements and their fit)

FIRST TEN MINUTES

1 SUPPLY ON LOAD

Power in (9V DC) pin 2 to pin 1; C50 positive side

+9 V ±2 V DC (7 to 11 V). From a 9 V bench supply a 48K draws 700-900 mA, a 16K 500-700 mA (3.1.4).

If not: Below +7 V the 7805 does not regulate: try another pack.

ZX Spectrum Service Manual 5.3.1 Basic checks, p. 26

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2 +5 V FROM THE 7805

C34 positive side; ULA pin 14; Z80A pin 11

+4.75 to +5.25 V, no visible ripple.

If not: Check the drop across R64, the 7805 output and input, and that the PWR socket is not shorted (6.2.3).

ZX Spectrum Service Manual 5.3.1 and 6.2.3, pp. 26 and 31

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3 -5 V FROM THE DC-DC CONVERTER

4116 pin 1

-4 V to -5.5 V.

If not: No -5 V: check TR4 and TR5, and change them together if TR4 blows again (6.1). Get -5 V back before fitting new 4116s.

ZX Spectrum Service Manual 3.1.2 Voltage check, p. 17

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4 +12 V FROM THE DC-DC CONVERTER

4116 pin 8

+12 V; investigate if more than 1 V out (outside 11 to 13 V).

If not: Keep powered checks short: new TR4/TR5 can burn out again. More than 1 V out after new TR4/TR5: check the converter diodes, the 5V1 zener and the 15 ohm resistors under the board, then lift 4116s one at a time; one shorted chip can pull a rail down.

Retroleum, Repairing a ZX Spectrum (lower RAM voltages, TR4/TR5)

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5 14 MHZ CRYSTAL

ULA pin 39

About 14 MHz.

If not: Check crystal X1, then the ULA (6.3.2).

ZX Spectrum Service Manual 6.3.2 No 3 MHz at IC2 pin 6, p. 32

6 CPU CLOCK *(unverified)*

ULA pin 32 to Z80A pin 6

3.5 MHz (WoS 48K reference) square wave with a full 0-5 V swing at Z80A pin 6.

If not: Low amplitude: R25, then TR3. Nothing at ULA pin 32 with 14 MHz present: ULA. The manual also asks for +3 V at ULA pin 13 (6.3.2).

ZX Spectrum Service Manual 6.3.2 No 3 MHz at IC2 pin 6, p. 32

7 RESET AND IDLE CONTROL LINES *(unverified)*

Z80A pin 26; Z80A pins 17, 24, 25

/RESET LOW at power-on, then HIGH. /NMI, /WAIT and /BUSRQ HIGH.

If not: A line held LOW: unplug everything from the edge connector, then compare each line at the Z80A and at its source with a good board (5.3.1.1).

ZX Spectrum Service Manual 5.3.1.1 Z80A control lines, p. 26

8 FRAME INTERRUPT AND LOWER RAM STROBES *(unverified)*

ULA pin 12; ULA pins 35, 1

/INT a short LOW pulse every 20 ms; /RAS and /CAS pulsing.

If not: All present but no boot: isolate the upper 32K by lifting +5 V from IC25 pin 16, then change IC13 down to IC6, then the ULA, then the Z80A, powering up after each change (5.3.1).

ZX Spectrum Service Manual 5.3.1 RAM isolation and IC change order, p. 27

PROBE HINTS

ULA.1 /CAS	Pulses LOW on every lower-RAM access.
ULA.4 /WE	Pulses LOW on writes to the lower RAM.
ULA.12 /INT	Pulses LOW every 20 ms (50 Hz).
ULA.13 +5V	+4.75 to +5.25 V *(unverified)* **Sources disagree:** ZX Spectrum Service Manual 6.3.2 and 6.4.7 (p. 32) say to check IC1 pin 13 for +3 V, not +5 V.
ULA.14 +5V	+4.75 to +5.25 V
ULA.28 SOUND	Toggles during BEEP, SAVE and LOAD.
ULA.32 CLK	3.5 MHz square wave, with gaps during the display.
ULA.33 /IOULA	Pulses LOW on ULA port access.